Project Summary and Planning Chart — Final Updated DRAFT

> **DRAFT — pending student / tutor review. Do NOT submit as final.**
This draft preserves the original form structure and identity fields, and updates only the fields supported by the Wave-1 evidence pack **plus the one-time Test/Holdout evaluation.** Original: `58324980_CHAN_Sing_Yeung_Project_Summary_and_Planning_Chart_202627A`. Supersedes the Wave-1 draft.

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1. Student and course identifiers

(RETAIN unchanged — copy from the submitted original.)

Field	Value
Student ID	58324980
Student name	CHAN Sing Yeung
Course / programme	[RETAIN original]
Academic year / term	2026/27 A
Supervisor	[RETAIN original]

2. Project title

[RETAIN original title; if it over-claims LLM superiority, add the qualifier below.]

> Qualifier (recommended): "…with explicit reporting of Non-LLM baselines, the seasonal-naive reference, and a held-out Test/Holdout evaluation."

3. Problem statement

(UPDATE)

It is unclear whether integrating frozen large language models into time-series forecasters yields accuracy that is (a) superior to lightweight linear models, (b) attributable to the text/LLM mechanism rather than to the underlying seasonality of influenza-like-illness (ILI) data, and (c) robust when evaluated on a held-out, distribution-shifted (COVID-era) window.

4. Research question and objectives

(UPDATE)

Research question: Across four Non-LLM and four LLM-based forecasting architectures, which performs best under the frozen selection rule; does the LLM family as a whole reliably outperform the Non-LLM family; and do the validation conclusions survive a one-time held-out Test/Holdout evaluation?

Objectives:

1. Run a pre-registered, seed-replicated (10 seeds × 4 horizons) benchmark of 8 architectures.
2. Identify best Non-LLM, best LLM, and overall Wave-1 validation champion under the frozen rule.
3. Ablate the LLM mechanism (LoRA, multimodal fusion, prompt components) to localize any text contribution.
4. Execute a one-time, audited Test/Holdout evaluation of the frozen champion and runner-up against a seasonal baseline, after a full freeze audit.
5. Preserve strict test/holdout isolation and a quota-safe compact-checkpoint pipeline.

5. Completed methodology

(UPDATE)

Item	Value
Dataset	National ILI, seven variables, z-space (train-only normalization)
Horizons	24, 36, 48, 60
Seeds	42 – 51 (10 frozen)
Early stopping	patience 7
Selection rule	rank by **average_MSE_4L** (mean of per-horizon mean MSE, lower better); secondary **average_MAE_4L**
Wave-1 execution	serial fail-stop controller, 580 accepted, 0 ledger FAIL
Test/Holdout execution	one-time, 84 runs, 25/25 terminal audit passed, frozen script hash unchanged

6. Models compared

(UPDATE)

Family	Models
Non-LLM	nlinear, dlinear, patchtst (PatchTST-inspired), itransformer
LLM-based	gpt2_miflu, qwen_time_llm, calf_llm, timecma_llm
Deterministic references	seasonal_naive, last_value_naive
Test/Holdout arms	gpt2_miflu, nlinear (10 seeds × 4 horizons), seasonal_naive (4 horizons)

7. Validated experimental scale

(UPDATE)

580 accepted rows = 320 primary + 8 baseline + 240 ablation + 12 negative control; 0 deleted / 0 duplicated. **Test/Holdout: 84 runs** = 2 stochastic models × 4 horizons × 10 seeds + 1 deterministic baseline × 4 horizons; 0 failed / 0 resubmitted.

8. Current progress and best result

(REPLACE)

Wave-1 (validation):

Result	Model	Value
Best Non-LLM	nlinear	avg MSE 4L = 1.0698 (95% CI [1.0677, 1.0719])
Best LLM	gpt2_miflu	avg MSE 4L = 0.7709 (95% CI [0.7200, 0.8217])
Overall Wave-1 validation champion	gpt2_miflu	runner-up nlinear (paired t = -13.33, $p = 3.13\times10^{-7}$)
Family finding	Non-LLM (1.142) vs LLM (1.243)	$p = 1.44\times10^{-6}$

Final (one-time Test/Holdout, COVID-era window):

Result	Model	Value
Best Test model	**nlinear**	avg MSE 4L = 2.739 (95% CI [2.731, 2.747])
Wave-1 champion on Test	gpt2_miflu	avg MSE 4L = 3.051 (95% CI [2.922, 3.179])
Seasonal baseline on Test	seasonal_naive	avg MSE 4L = 4.279 (deterministic)
Champion generalizes?	—	**No** — ordering reverses (C1: paired $t = 5.46$, $p = 4.0\times10^{-4}$)
Learned models beat baseline?	—	**Yes** — C2 $p = 4.6\times10^{-9}$, C3 $p = 7.7\times10^{-21}$
Peak intensity (Test, PIE)	nlinear	9–21% (vs gpt2 41–62%, seasonal 50–124%)

Central finding: the LLM champion's validation advantage was window-specific; on a held-out, distribution-shifted window the simple linear **nlinear** is the more robust and more accurate model.

9. Contributions

(REPLACE — conservative, evidence-supported)

1. Empirical architecture comparison — pre-registered, seed-replicated, test-isolated benchmark of 8 Non-LLM/LLM architectures under a frozen selection rule.
2. Held-out generalization evidence — a one-time, audited Test/Holdout showing the validation champion's advantage does not survive distribution shift, while both learned models still beat the seasonal baseline.
3. Mechanism / ablation evidence — the LLM text contribution localized to the dataset-information component; LoRA and multimodal fusion statistically inert.
4. Reproducible quota-safe framework — fail-closed compact-checkpoint pipeline (580/580 + 84/84 accepted, 0 fail events).

10. Ablation and control work

(UPDATE)

6 ablations: only **without_dataset_information** significant (+12.85%, $p = 0.004$); LoRA / multimodality / task-instruction not significant. 12 executed negative controls (exploratory): zero_text −0.063, shuffled_text +0.081. 18 deferred negative controls remain pending Wave-2 (parent binding = **gpt2_miflu**).

11. Remaining work

(REPLACE)

Final report written (EN/ZH supervisor report + tutor brief + this summary). Remaining: student/tutor review and sign-off; manuscript drafting; final QA manifest; decision on whether to run the 18 deferred Wave-2 controls before manuscript finalization.

12. Revised timeline

(REPLACE)

1. ~~Wave-2 deferred-control binding + execution~~ (pending, if required for manuscript).

2. ~~Freeze audit + final champion selection~~ → freeze audit passed; selection frozen at `gpt2_miflu`.
3. ~~Test/holdout evaluation only after freeze~~ → **complete** (84/84, 25/25 audit).
4. ~~Final metrics / figures / reports~~ → **complete** (this report set).
5. Manuscript drafting and submission (remaining).

13. Risks and mitigations

(REPLACE)

Risk	Mitigation
LLM superiority over-claimed	report family-level result + seasonal_naive honestly; Test reversal now documented
Validation advantage does not generalize (REALIZED)	reported plainly: Test ordering reverses, **nlinear** best on held-out window
gpt2_miflu seed instability (SD 0.071)	report seed SD + 95% CI
Distribution shift conflated with COVID	state explicitly the Test window is a single contiguous COVID-era window
PTE/PIE (peak metrics)	secondary metric, never a tie-breaker; documented lead-1 + reconstructed-target convention
patchtst fidelity (no pretraining/RevIN)	carry "PatchTST-inspired" caveat
LLM cost (409× params, 28× time vs nlinear)	present cost table transparently; now reinforced by Test result

14. Status labels

(ADD — must appear verbatim)

- Wave 1 complete
- Test/Holdout evaluated (one-time, 84/84, 25/25 audit)
- final validation champion: gpt2_miflu (frozen); final Test best: nlinear
- 18 deferred controls pending binding/execution
- final report/manuscript pending student & tutor sign-off